Program : **Diploma in Tool & Die Engineering**	
Course Code : **3119**	Course Title: **3D Modelling Lab**
Semester : **3**	Credits: **No Credit**
Course Category: **Program Core**	
Periods per week: **4 (L:0, T:0, P:4)**	Periods per semester: **60**

Course Objectives:

- To familiarize students with the basics of Solid works/Inventor/Creo/NX CAD /Delcam designing software. (Any one of the given software)
- To develop the ability of students to make production drawing using software.
- To impart the basic concepts of 3D modeling.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge about all aspects of drawing, Orthographic projection		Engineering Graphics	1
CAD software basic tools		Basic CAD Lab	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply the basic concepts of 3D modeling in a software environment.	16	Applying
CO2	Sketch various fastening devices by choosing proper tools in the software.	12	Applying
CO3	Prepare detailed drawing of a complex component in a fast and effective manner.	21	Applying
CO 4	Interpret GD&T symbols and surface finish symbols in a drawing.	8	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			3
CO4	3						3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Apply the basic concepts of 3D modeling in a software environment.**		
M1.01	Identify the use of different 3D tools and commands	6	Understanding
M1.02	Prepare different 3D modelling of simple objects	10	Applying
Contents: Introduction to CAD, CAE, PDM Features of SolidWorks, Various products available in SolidWorks for Product Design, Simulation, Communication SolidWorks Graphical User Interface - Feature manager design tree, Callouts, Handles, Confirmation corner, mouse buttons, keyboard shortcuts, Command Manager, Hardware and Software requirements, SolidWorks Task Scheduler, SolidWorks Rx Sketcher Sketch Entities - Inference line, Centerline line, Line, Circle, Arc, Ellipse, Rectangle, Slots, Polygon, Parabola, Ellipse, Partial Ellipse, Spline, Spline tools, Spline on surface, Equation driven curve, Points, Text, Construction geometry, Snap, grid Sketch Tools - Fillet, Chamfer, Offset, Convert entities, Intersection curve, Face curve, Trim, Extend, Split, Jog Line, Construction Geometry, Mirror, Dynamic Mirror, Move, Copy, Rotate, Scale, Stretch, Sketch pattern , Polygon, Make path, Close Sketch To Model, Sketch picture, Check Sketch for Feature, Area hatch/Fill Blocks Make block, edit block, insert block, Add/Remove Entities, Rebuild, Save, Explode Relations - Adding Sketch Relation, Automatic relations, Dimensioning - Smart, Horizontal, Vertical, Ordinate, Horizontal ordinate, Vertical ordinate, align ordinate, fully define sketch. Sketch Diagnosis, Sketch Xpert, 3D Sketching, Rapid Sketch			

CO2	**Sketch various fastening devices by choosing proper tools in the software.**		
M2.01	Recall concepts of fastening devices.	2	Remembering
M2.02	Sketch a diagram using Solidworks tools.	10	Applying
	Lab Exam - I	1.5	

Contents:

Part Modelling

Part Modeling Tools

Creating reference planes

Creating Extrude features - Direction1, Direction2, From option, Thin feature, applying draft, Selecting contours

Creating Revolve features - Selecting Axis, Thin features, Selecting contours

Creating Swept Features-Selecting, Profile and Path, Orientation/twist type, Path Alignment, Guide Curves, Start/End tangency, Thin feature

Creating Loft features - Selecting Profiles, Guide curves, Start/End Constraints, Centerline parameters, Sketch tools, Close loft.

Selecting geometries - Selection Manager, Multiple Body concepts

Creating Reference - points, axis, coordinates

Creating curves -

Split curve, Project curve, Composite curve, Curve through points, Helix and Spiral

Creating Fillet features Inserting Hole types

Creating Chamfer Creating Shell Creating Rib

Creating Pattern - Linear pattern, Circular pattern, Sketch driven pattern, Curve driven pattern, Table driven pattern, Fill pattern, mirror Advanced Modeling Tools- Dome, Free form, Shape feature,Deform, indent, Flex

Exercises in 3D Modelling of the following.

Draw assembly of a bolt, nut and a plain washer (Hexagonal and Square headed),

Draw rivet heads (general purpose) and riveted joints using standard proportions. Single riveted and double riveted lap joint (Chain and zigzag), single riveted single strap butt joint and single riveted double strap butt joint.

CO3	**Prepare detailed drawing of a complex component in a fast and effective manner**		
M3.01	Recall construction details of components and bill of materials required for cotter joints, couplings, bearings and machine parts.	1	Remembering
M3.02	Prepare detailed drawings in Solid works	20	Applying

Contents:

Assembly Modelling

Assembly Modeling Tools

Introduction to Assembly Modeling & Approaches – Top down and Bottom-up approach

Applying Standard Mates- Coincident, Parallel, Perpendicular, Tangent, Concentric, Lock, Distance, Angle.

Surface Modelling

Surface Modeling tools

Creating Extrude, Revolve, Swept, loft, Boundary surface. Inserting Planar Surface, Offset Surface, Radiate Surface.

Extending a surface, Surface fill, Ruled Surface, Trimming Surface, Mid surface, Replace Face, delete face, Un trim surface, Knit surface, Thickening a Surface, Move Face

Detailed drawings of following machine parts are to be made by applying different commands in Solid work.

1) Sleeve & Cotter Joint

2) Socket & Spigot Joint

3) Stuffing Box

4) Foot Step bearing or Plummer Block

5) Flanged Coupling Unprotected type

6) Plastic bottle surface modeling

CO4	**Implement GD&T symbols and surface finish symbols in a drawing.**		
M4.01	Recall the concept of GD&T and surface finish	2	Remembering
M4.02	Integrate CAD skill and shop floor drawing knowledge.	6	Applying
	Lab Exam - II	1.5	

Contents:

Drafting

Generating Drawing Views Introduction to Angle of Projection

Generating Views - Generating Model View, Projected Views, Inserting Standard 3 View

View creation relative to model, inserting predefined views, empty views, Auxiliary Views, Detailed Views, Crop view, broken –Out Section, Broken Views, Section View, Aligned Section View, Alternate Position View, working assembly specific view, drawing properties, Manipulating views

Exercise GD&T in AutoCAD/Solidworks Surface finish symbols using BLOCK

Prepare shop floor drawing for the Following 1) Slip Bush 2) Over Hung Crank 3) C- Clamp 4) Connecting Rod

Text/ Reference:

T/R	Book Title/Author
T1	P I Varghese, K C John., Machine Drawing, VIP Publishers
T2	K C John, Textbook of Machine Drawing, PHI Learning Private Limited, New Delhi
R1	Bhatt, N.D., Machine Drawing, Charotar Publishing House, 2003.
R2	PS Gill, Machine Drawing, Katson Publishing.

Online Resources:

Sl.No	Website Link
1	http://www.nptelvideos.in/2012/12/computer-aided-design.html
2	https://knowledge.autodesk.com/
3	https://www.thesourcecad.com/autocad-tutorials/